SABRE+ TIFS Best-Paper-Level Experiment Roadmap

A detailed plan to strengthen Extended Agentic SABRE for IEEE Transactions on Information Forensics and Security submission

Focus: ransomware detection, uncertainty-aware multi-agent fusion, knowledge augmentation, continual adaptation, robust security evaluation

Bottom line: the manuscript already has a strong idea - a three-agent, uncertainty-aware, knowledge-augmented detector. To compete at TIFS best-paper level, the evidence must shift from "promising prototype" to "credible, stress-tested, reproducible system". The highest-impact additions are a unified tri-modal benchmark, family/time OOD tests, empirical continual-learning experiments, stronger uncertainty mathematics, adversarial robustness evaluation, and statistically rigorous comparisons.

Safety note: all malware/ransomware experiments in this plan assume isolated laboratory infrastructure, benign emulation where possible, strict institutional approval, no public release of weaponized samples, and reporting focused on defensive measurement rather than operational attack guidance.

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1. Diagnosis of the Current Manuscript

The current paper presents Extended Agentic SABRE (SABRE+), with a Semantic CNN, Behavioural CNN, and Knowledge CNN, each producing mean probability and MC Dropout uncertainty. It adds peer consultation, Bayesian threshold optimization, a triage policy, CTGAN score augmentation, and an expert feedback loop. The idea is publishable; the main risk is that the current evidence may look under-validated for a top archival venue.

Type	Observation
Strong contribution	Three-agent architecture: semantic PE metadata, behavioral runtime telemetry, and practitioner knowledge. This is a clear systems contribution beyond a single classifier.
Strong contribution	Uncertainty-aware triage: AUTO-CONTAIN requires high risk and low uncertainty, while uncertainty can trigger escalation. This maps naturally to real SOC workflows.
Strong contribution	Knowledge Agent: RansKnow brings MITRE ATT&CK/tool/platform features into the detector rather than treating threat intelligence as prose.
Major blocker	No unified tri-modal dataset: the paper says RDset, RanSMAP, and RansKnow are independent, so fusion evaluation cannot prove all three modalities improve the same samples.
Major blocker	RansKnow is small and heuristically labeled: 307 samples with rule-based labels is a vulnerability reviewers will attack.
Major blocker	Semantic AUC = 1.0 may be interpreted as dataset saturation or leakage unless there are family/time/producer holdouts and leakage audits.
Major blocker	Expert Feedback Loop is formalized but not empirically evaluated in a longitudinal stream.
Major blocker	External baselines appear to rely partly on literature values rather than reimplementation under the same splits.
Major blocker	No formal statistical significance analysis, uncertainty calibration stress tests, deployment latency/cost analysis, or adversarial evasion benchmark.

Immediate reviewer concern to preempt

Reviewers will likely ask: does SABRE+ truly beat two-agent SABRE under realistic unseen ransomware families and realistic benign enterprise base rates, or is the improvement an artifact of independent datasets and strong semantic separation? The experiment plan below is designed to answer that question decisively.

2. What "Best Paper Level" Means Operationally

No experiment can prove a system is the "best solution ever" in cybersecurity. Cybersecurity is adversarial and changes over time. A best-paper-level claim is more defensible: SABRE+ should be the most rigorously evaluated, mathematically principled, and deployable uncertainty-aware multi-agent ransomware detection framework demonstrated to date.

Criterion	What SABRE+ needs to show
Technical depth	Add formal decision theory, conformal risk control, OOD statistics, sequential detection, continual-learning analysis, and uncertainty decomposition.
Originality	Move from a CNN ensemble to a principled agentic system: uncertainty-gated peer consultation, knowledge graph augmentation, and analyst feedback under drift.
Impact	Demonstrate lower analyst workload at equal or better recall under realistic imbalance, unseen families, temporal drift, and adversarial/mimicry settings.
Reproducibility	Release code, split definitions, feature extraction scripts, calibration recipes, model cards, dataset cards, and a non-sensitive synthetic/derived benchmark where possible.
Security realism	Use family/time splits, malware-lab telemetry, benign enterprise workloads, false-positive cost, latency, and safe adversarial emulation.

3. Experimental Master Plan

The master plan is organized into eight research questions (RQs). Each RQ should become either a table, figure, or subsection in the revised paper.

- RQ1 - Tri-modal value: Does semantic + behavioral + knowledge fusion outperform every single agent and every two-agent combination on the same samples?
- RQ2 - OOD resilience: Does SABRE+ maintain recall, calibration, and controlled escalation on unseen ransomware families, unseen campaigns, and future time periods?

- RQ3 - Uncertainty correctness: Are high uncertainty and high sigma_max associated with higher error likelihood, OOD samples, label noise, and adversarial perturbation?
- RQ4 - Peer consultation: Does consultation improve high-uncertainty decisions without harming calibration or over-amplifying wrong confident agents?
- RQ5 - Continual adaptation: Does the Expert Feedback Loop reduce degradation under drift while avoiding catastrophic forgetting?
- RQ6 - Knowledge robustness: Does RansKnow improve when expanded from YouTube transcripts to STIX/TAXII, ATT&CK technique-level, and expert-labeled intelligence?
- RQ7 - Adversarial realism: How much work must an adversary do to evade semantic, behavioral, and knowledge channels simultaneously?
- RQ8 - SOC deployability: What are latency, memory, throughput, false-positive cost, escalation volume, and analyst utility?

ID	Experiment	Protocol	Primary outputs	Priority
E1	Unified tri-modal benchmark	Build matched samples with PE/static, sandbox/runtime, and family/campaign/ATT&CK intelligence for the same executable/campaign.	Dataset schema; leakage audit; train/val/test splits	Mandatory
E2	Family-level holdout	Train on known families; test on fully unseen families and campaigns.	AUROC, AUPRC, balanced accuracy, recall@FPR, ECE, escalation rate	Mandatory
E3	Temporal holdout	Train on older samples; validate on middle period; test on future samples.	Performance decay, calibration drift, detection delay	Mandatory
E4	Realistic imbalance	Evaluate at enterprise-like base rates and benign workloads; report PR-AUC and cost.	AUPRC, false positives/day, cost-weighted risk	Mandatory
E5	Full ablation	Compare all 7 modality subsets plus no-consult/no-knowledge/no-uncertainty variants.	Delta metrics with 95% CIs and significance tests	Mandatory
E6	Uncertainty benchmark	MC Dropout vs deep ensembles, Laplace approximation, evidential DL, temperature scaling, Dirichlet calibration.	NLL, Brier, ECE, AUROC-error, risk-coverage	High
E7	Conformal triage	Wrap fused risk with conformal prediction to guarantee error/coverage on held-out calibration data.	Coverage, selective risk, escalation fraction	High
E8	Peer consultation variants	Inverse-std, inverse-variance, product-of-experts, mixture-of-experts gating, attention message passing, Dempster-Shafer.	Accuracy on high-sigma samples, calibration damage, failure cases	High
E9	Knowledge source expansion	Add STIX/TAXII, ATT&CK technique descriptions, vendor reports, CISA advisories, and expert annotation.	Knowledge-agent AUC, label-noise sensitivity, zero-shot family lift	High
E10	Expert feedback simulation	Replay a chronological stream with delayed analyst labels and bounded budgets.	Prequential AUC, forgetting, recovery time, update latency	Mandatory
E11	Active learning	Select escalations by uncertainty, expected model change, BALD, expected cost reduction.	Accuracy per analyst label; workload reduction	High
E12	Adversarial/mimicry stress	Feature-space evasions and safe malware-lab emulation: static packing/metadata shifts, slow encryption, LotL-like telemetry.	Attack success rate, required perturbation, escalation safety	Mandatory
E13	Poisoning/backdoor stress	Simulate mislabeled feedback and contaminated replay memory; test robust updates.	Attack success, robust training defense, rollback detection	High
E14	Synthetic data audit	Assess CTGAN usefulness and	Nearest-neighbor leakage, TSTR/TRTS, high-sigma	High

		privacy/leakage; compare to SMOTE and no augmentation.	coverage	
E15	Explainability validation	Integrated gradients/SHAP/counterfactuals; analyst study on explanations.	Fidelity, stability, analyst precision/time/trust	Medium
E16	Runtime/deployment benchmark	Measure end-to-end latency, memory, throughput, GPU/CPU cost, and triage queue impact.	p50/p95 latency, samples/sec, cost per endpoint/day	Mandatory

4. Mathematical Techniques to Add Depth and Novelty

The current manuscript already uses MC Dropout, inverse-uncertainty peer weighting, Bayesian threshold optimization, temperature calibration, and replay. The following additions would make the mathematics more convincing and more TIFS-ready.

Technique	Core formula or protocol	Why it matters
Decision-theoretic triage	Define actions A = {allow, escalate, contain} and choose a*(x)=argmin_a E[C(a,Y)|x]. Costs include false contain, missed ransomware, analyst escalation, and delay.	Turns thresholds into a principled security-economics policy.
Conformal prediction	Use nonconformity score S(x,y)=1-p_y(x). Calibrate q_(1-alpha). Escalate when the conformal set is ambiguous or excludes safe decisions.	Provides finite-sample risk/coverage guarantees under exchangeability.
Selective classification	Report risk-coverage curve: R(g)=E[L(f(X),Y)g(X)] / E[g(X)], where g(X)=1 means autonomous decision.	Proves uncertainty reduces unsafe automation rather than only improving accuracy.
Uncertainty decomposition	For T stochastic predictions, decompose predictive entropy into aleatoric and epistemic components using mutual information.	Shows whether uncertainty is model ignorance, data ambiguity, or label noise.
Bayesian hierarchical evaluation	Model logit(metric) with random effects for dataset, family, time window, and seed. Report posterior P(SABRE+ > baseline).	Stronger than a single average table; handles correlated malware-family results.
Optimal transport/domain shift	Compute Wasserstein distance between train/test score distributions and correlate shift with performance decay.	Quantifies distribution shift instead of only naming concept drift.
MMD two-sample tests	MMD^2(P,Q)=||E_P phi(X)-E_Q phi(X)||^2. Use it on embeddings and score vectors across time/families.	Detects when validation assumptions fail.
Sequential detection	Use CUSUM/SPRT on behavioral streams rather than only static sample classification.	Fits ransomware behavior: early detection before encryption completes.
Graph threat intelligence	Build a graph G=(V,E) with families, tools, ATT&CK tactics/techniques, platforms, IOCs, and campaigns; use GNN/message passing.	Raises Knowledge Agent from count features to relational reasoning.
Causal/invariant learning	Use invariant risk minimization or anchor regression over environments: family, time, sandbox, source.	Finds features stable across changing malware ecosystems.
Robust continual learning	Compare replay with EWC, SI, LwF, GEM/AGEM, ER-ACE; report forgetting F_i=max_t a_i,t - a_i,T.	Empirically validates the Expert Feedback Loop.
Multiple-test control	Use DeLong for AUC, McNemar for paired classification, bootstrap CIs, and Holm-Bonferroni correction.	Prevents overclaiming across many datasets and variants.

Expected cost: R(a|x) = c_FN P(Y=1|x) 1[a=allow] + c_FP P(Y=0|x) 1[a=contain] + c_ESC 1[a=escalate] + c_LAT delay(a)

Conformal set: Gamma_alpha(x) = { y in {0,1}: S(x,y) <= q_(1-alpha) }. Escalate if |Gamma_alpha(x)| != 1 or if the singleton action is too costly.

Risk-coverage: coverage = E[g(X)], selective_risk = E[L(f(X),Y) g(X)] / E[g(X)].

MMD shift: MMD^2(P,Q) = || E_P[phi(X)] - E_Q[phi(X)] ||_H^2.

Continual forgetting: F_i = max_{t<T} A_{i,t} - A_{i,T}; average forgetting = mean_i F_i.

5. Detailed Experiment Protocols

5.1 Build a unified tri-modal benchmark

This is the single most important experiment. Without matched modalities, the fusion results can be criticized as not truly end-to-end. Create a benchmark where each malware/campaign record links to: (a) PE/static features, (b) sandbox or EDR runtime telemetry, (c) knowledge features from ATT&CK/STIX/TAXII/practitioner intelligence, and (d) family/campaign/time labels.

- Unit of analysis: executable hash when available; otherwise campaign-family-time tuple. Keep a clear distinction between sample-level and campaign-level knowledge.
- Minimum target: 5,000 to 20,000 malware samples, 50,000+ benign executables/telemetry windows, 20+ ransomware families, and at least 12 months of timestamps. Smaller is acceptable if high quality and fully documented.
- Benign data: include enterprise software updates, backup software, compression/encryption tools, developer tools, database workloads, VSS/backup operations, and administrative scripts that are false-positive traps.
- Runtime data: execute only in an isolated malware lab or use benign emulation. Capture file rename/write entropy, directory traversal, process tree, registry, network, memory, API calls, and EDR-like event streams.
- Knowledge data: map family/campaign to ATT&CK tactics and techniques, tools, platforms, C2/exfiltration, and incident reports. Store provenance and timestamp for each intelligence source.
- Leakage audit: split by family, campaign, signer, packer, source, time, and near-duplicate hashes. Report deduplication method and hash/embedding similarity thresholds.

Field	Purpose
sample_id	Stable internal ID; never use path or family name as model feature.
sha256 / fuzzy hash	Used for deduplication and provenance; not used as a model feature.
family, campaign, collection_date	Required for family and temporal holdouts.
static_features	PE headers, imports, section entropy, strings, byte/image embeddings.
behavioral_features	EDR/sandbox windows: file ops, entropy, process tree, registry, memory, network.
knowledge_features	ATT&CK tactics/techniques, tools, platforms, STIX entities, source confidence.
label and label_source	Ransomware/benign, expert-confirmed vs heuristic vs weak label.
safety_metadata	Lab environment, detonation controls, sample handling restrictions.

5.2 Family/time/generalization evaluation

Run four split types. All models must use the same split definitions.

Split	Definition	Main claim supported
Random stratified	For continuity with current paper only. Do not make it the headline result.	Should be highest performance.
Family holdout	Leave out complete ransomware families during training.	Measures unseen-family generalization.
Temporal holdout	Train on months 1-k, validate on k+1, test on k+2 onward.	Measures concept drift and future realism.
Source/lab holdout	Train on one collection/sandbox/source, test on another.	Measures collection and instrumentation bias.
Benign hard-negative holdout	Exclude a benign category such as backup/encryption tools from training and test on it.	Measures false-positive realism.

5.3 Full ablation and contribution proof

A best-paper-level systems claim requires showing exactly which component matters. Report each ablation under random, family, temporal, and imbalanced settings.

- Agents: semantic only, behavioral only, knowledge only, semantic+behavioral, semantic+knowledge, behavioral+knowledge, all three.
- Fusion: simple average, logistic regression, calibrated stacking, current Decision MLP, product-of-experts, mixture-of-experts gate.
- Uncertainty: no uncertainty features, sigma features only, sigma_max triage, conformal triage, MC Dropout T sensitivity.
- Peer consultation: off, current inverse-uncertainty, learned attention gate, robust consultation that ignores disagreeing overconfident peers.
- Knowledge: current RansKnow counts, expert-labeled RansKnow, ATT&CK technique-level, STIX/TAXII graph, LLM-extracted features with human verification.

- Continual adaptation: no update, fine-tune MLP only, replay, EWC, replay+EWC, active learning budget.

5.4 Uncertainty and calibration experiments

Treat uncertainty as a first-class security property. Do not only report ECE on balanced splits.

Item	Definition/use
ECE and adaptive ECE	Reliability across probability bins; adaptive bins avoid empty-bin artifacts.
Brier score and NLL	Proper scoring rules for probability quality.
AUROC-error	Use sigma or entropy to predict whether the model is wrong.
AURC/E-AURC	Area under risk-coverage curve for selective classification.
OOD detection AUROC	Can uncertainty separate unseen families, future samples, or unseen benign workloads?
Calibration under attack	Does evasion increase uncertainty and route to escalation instead of allow?

5.5 Empirical Expert Feedback Loop

Turn the formal feedback loop into a longitudinal experiment. Simulate deployment with delayed labels and analyst budgets.

- Create a chronological stream of samples. At each time step, make a triage decision using the current Decision MLP.
- Only escalated and auto-contained samples can receive analyst labels. Add a delayed-label variant: labels arrive after 1, 7, or 30 days.
- Budget constraints: 10, 25, 50, and 100 labels/day. Compare uncertainty sampling, random escalation sampling, expected-cost sampling, and BALD.
- Update schedules: daily, weekly, drift-triggered by MMD/KS shift, and emergency update after a spike in uncertainty.
- Baselines: no update, naive fine-tune, replay, EWC, LwF, GEM/AGEM, replay+EWC.
- Metrics: prequential AUC/AUPRC, false negatives during drift, false positives/day, average forgetting, recovery time after new family appears, update latency, and calibration after update.

5.6 Adversarial and robustness evaluation

Keep this defensive and feature-level unless you have a fully approved malware lab. The goal is to quantify robustness, not to publish evasion recipes.

Stress test	Safe protocol	Question answered
Static semantic evasion	Feature-space PE metadata perturbations, packer/source holdout, signer/source leakage audit.	Does sigma_sem rise and route to escalation?
Behavioral mimicry	Safe emulation of slow encryption, throttled writes, backup-like behavior, renamed processes, LotL-like process trees.	Detect early enough before damage threshold?
Knowledge evasion	Unseen family with sparse public intelligence; contradictory intelligence; missing ATT&CK mappings.	Does system avoid over-trusting weak knowledge?
Multi-modal evasion	Combine plausible feature shifts across two modalities while keeping third intact.	How much adversary effort is needed to reach ALLOW?
Poisoned feedback	Inject mislabeled analyst corrections or adversarial replay samples.	Can robust update or audit detect and contain poisoning?
Data augmentation leakage	Membership/nearest-neighbor and train-on-synthetic-test-on-real checks for CTGAN.	Does CTGAN help without memorizing?

6. Statistical Analysis and Reporting Package

Top-tier reviewers expect statistical rigor. Report uncertainty over results, not only model uncertainty.

- Repeat each training run with at least 5 random seeds; use fixed published splits for main results.
- Report 95% bootstrap confidence intervals for AUROC, AUPRC, recall@FPR, F1, ECE, escalation rate, and cost.
- Use DeLong tests for paired AUROC comparisons; McNemar or exact tests for paired classification differences; paired bootstrap for AUPRC and escalation rate.
- Apply Holm-Bonferroni or Benjamini-Hochberg correction across ablations to avoid false discoveries.
- Report effect sizes, not only p-values: absolute delta, relative error reduction, and analyst-hours saved.
- Use hierarchical Bayesian analysis for family-level results so one large family cannot dominate conclusions.
- Pre-register the primary claim: e.g., "SABRE+ improves family-holdout AUPRC and reduces analyst escalation at equal recall compared with two-agent SABRE."

Claim	Primary metric	Test/interval	Minimum convincing result
Tri-modal beats two-agent	AUPRC and recall@FPR on family/time holdout	Paired bootstrap + DeLong where applicable	Positive delta with CI excluding 0 on at least two hard splits
Uncertainty helps safety	Selective risk at 80/90/95% coverage	Bootstrap CI	Lower autonomous error at same coverage
Consultation helps	High-sigma subset AUPRC/ECE	Paired bootstrap	Improves high-sigma accuracy without ECE degradation
Feedback adapts	Prequential AUPRC and forgetting	Hierarchical model over drift windows	Faster recovery and lower forgetting than naive fine-tune
Deployable	FP/day, p95 latency, memory	Bootstrap over days/endpoints	Meets target SOC workload and latency budget

7. Figures, Tables, and Manuscript Changes

New artifact	Purpose
Figure 1	Revised architecture with three agents, knowledge graph, conformal/decision layer, and feedback loop.
Figure 2	Unified tri-modal dataset schema and split strategy; show family/time/source separation.
Figure 3	Main performance: grouped bars or forest plot with 95% CIs across random, family, temporal, imbalance splits.
Figure 4	Risk-coverage curves showing autonomous decision safety versus escalation workload.
Figure 5	OOD uncertainty: sigma/entropy distributions for in-distribution, unseen family, future time, and adversarial/mimicry samples.
Figure 6	Continual-learning timeline: prequential AUPRC, false negatives, and recovery after new-family drift.
Figure 7	Peer consultation failure/success cases with calibration impact.
Figure 8	Threat-intelligence graph view: families-tools-ATT&CK-platforms with GNN attention or feature attribution.
Figure 9	Runtime/SOC cost curve: false negatives, false positives/day, escalation hours/day, latency.

Table	Contents
Table A	Dataset statistics by split: families, dates, benign categories, label sources, modality availability.
Table B	Leakage audit: deduplication rates, near duplicates removed, source separation.
Table C	Main comparison under identical splits: SABRE+, 2-agent SABRE, single-agent models, classical ML, recent deep baselines.
Table D	Ablation table for each component: Knowledge Agent, consultation, uncertainty, CTGAN, calibration, feedback.
Table E	Uncertainty method comparison: MC Dropout, ensembles, Laplace, evidential, conformal.
Table F	Continual learning baselines and forgetting.
Table G	Adversarial/mimicry robustness and escalation safety.
Table H	Deployment resources and latency.

8. Timeline and Execution Roadmap

A realistic schedule is 12 to 16 weeks if data collection can proceed immediately. The fastest path is to run the mandatory experiments first, while preparing the paper rewrite in parallel.

Time	Deliverable
Week 1	Audit current code/data; freeze baseline; define claims, metrics, and split files; create reproducibility checklist.
Weeks 2-3	Build/align unified tri-modal benchmark; deduplicate; create family/time/source/hard-negative splits.
Weeks 4-5	Run baseline reimplementations and full ablation: all modality subsets, no consultation, no uncertainty, no knowledge.
Weeks 6-7	Run OOD/family/time/imbalance experiments; compute calibration, selective risk, and statistical tests.
Weeks 8-9	Run uncertainty methods, conformal triage, and peer-consultation variants.
Weeks 10-11	Run longitudinal Expert Feedback Loop with delayed labels, active learning, and continual-learning baselines.
Weeks 12-13	Run safe adversarial/mimicry, poisoning, CTGAN audit, and deployment latency/cost experiments.
Weeks 14-15	Create figures/tables; write revised methods/results/limitations; release reproducibility artifacts.
Week 16	Internal red-team review: ask coauthors to attack leakage, claims, assumptions, and reproducibility; finalize submission.

9. Minimum Viable Package vs. Moonshot Package

9.1 Minimum viable package for a strong TIFS submission

- Unified tri-modal or at least partially matched dataset for a meaningful subset.
- Family-level and temporal holdout results; do not lead with random split performance.
- Full ablation proving Knowledge Agent and peer consultation add value beyond two-agent SABRE.
- Empirical Expert Feedback Loop with chronological stream, delayed labels, replay baseline, and forgetting metrics.
- Conformal or selective-classification analysis showing safe reduction in analyst workload.
- Adversarial/mimicry stress tests showing uncertainty routes uncertain evasions to escalation.
- Reimplemented baselines under identical splits, confidence intervals, and significance tests.
- Reproducibility package: code, split IDs, feature-extraction details, model checkpoints, and dataset card.

9.2 Moonshot package for best-paper-level impact

- Public SABRE-Triplet benchmark with static, behavioral, and knowledge modalities; a carefully sanitized subset if raw samples cannot be released.
- Knowledge graph agent using ATT&CK/STIX/TAXII with graph neural networks and provenance-aware confidence.
- Finite-sample risk guarantees via conformal triage and cost-sensitive decision theory.
- Cross-lab evaluation: train on one sandbox/telemetry source, test on another.
- Analyst-in-the-loop user study showing explanations and uncertainty reduce triage time without increasing missed ransomware.
- Robust feedback loop with poisoning detection and rollback under malicious/mistaken analyst labels.
- Open leaderboard-style evaluation scripts for future ransomware detectors.

10. References and Source Notes

Use the sources below to support the revised paper and the motivation for this experiment plan. The current manuscript is referenced as [R1].

[R1] User-provided manuscript: Extended Agentic SABRE: Knowledge-Augmented Three-Agent Ransomware Detection via Uncertainty-Driven Peer Consultation and Expert-Guided Continual Adaptation, SABRE_TIFS.pdf, 2026 draft.

[R2] IEEE Signal Processing Society, IEEE Transactions on Information Forensics and Security scope and reproducible research guidance. https://signalprocessingsociety.org/publications-resources/ieee-transactions-information-forensics-and-security/ieee-transactions

[R3] IEEE Signal Processing Society awards guidance. Judging basis includes technical depth, originality, and impact. https://signalprocessingsociety.org/community-involvement/awards-submit-award-nomination

[R4] IEEE Signal Processing Society 2025 Awards Winners, including TIFS papers "Noiseprint" and "VerifyNet" among Best Paper Awards. https://signalprocessingsociety.org/newsletter/2026/01/2025-ieee-signal-processing-society-awards-winners

[R5] MITRE ATT&CK. Globally accessible knowledge base of adversary tactics and techniques based on real-world observations. https://attack.mitre.org/

[R6] OASIS Cyber Threat Intelligence Technical Committee. STIX/TAXII documentation. https://oasis-open.github.io/cti-documentation/

[R7] CISA StopRansomware Guide. https://www.cisa.gov/resources-tools/resources/stopransomware-guide

[R8] NIST AI Risk Management Framework. https://airc.nist.gov/airmf-resources/airmf/

Appendix A. Reproducibility Checklist for the Revised Paper

- Publish immutable split files and random seeds.
- Document all feature extraction, preprocessing, scaling, imputation, and normalization steps.
- Separate threshold tuning, calibration, and testing data; never tune thresholds on the test set.
- Report training hardware, inference hardware, dropout passes T, latency, and energy/cost where possible.
- Publish model cards and dataset cards, including label sources, known biases, and unsafe-use restrictions.
- Include a leakage audit: near duplicates, family/source separation, timestamp leakage, label-feature leakage.
- Release non-sensitive knowledge features and derived score vectors if raw malware cannot be released.
- Provide scripts to reproduce every figure and table from a clean environment.